

# NICOLA BORGHI

## Curriculum Vitae

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From Renazzo (Ferrara), Italy, 22 May 1996  
Dipartimento di Fisica e Astronomia “Augusto Righi”  
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**Summary:** Postdoctoral researcher in gravitational-wave cosmology and galaxy evolution at the University of Bologna. Developer of two public codes to constrain the expansion history of the Universe using gravitational waves and galaxy catalogs (**CHIMERA**) and stellar population features in galaxy spectra (**PyLick**). Co-lead of Dark Sirens Work Package in Euclid and Cosmology-Additional Probes in WST. Member of the LVK, Einstein Telescope, Euclid, and WST collaborations. Passionate about science communication and coordinator of various outreach activities.

## Current position

2023–present **Postdoctoral researcher in Gravitational-wave Cosmology & Galaxy Evolution.**  
Department of Physics and Astronomy “Augusto Righi”, University of Bologna, Italy

## Education

2020–2023 **PhD, Astrophysics**, University of Bologna, Italy.  
Thesis: *Unveiling the Expansion History of the Universe with Cosmic Chronometers and Gravitational Waves*. [Link](#).  
Supervisors: Michele MORESCO & (co-) Andrea CIMATTI

2018–2020 **Master of Science, Astrophysics and Cosmology (cum laude)**, University of Bologna, Italy.  
Supervisors: Andrea CIMATTI & (co-) Michele MORESCO

2015–2018 **Bachelor of Science, Astronomy (cum laude)**, University of Bologna, Italy.

2010–2015 **Scientific High School Diploma**, ISIT Bassi Burgatti, Cento (FE), Italy.

## Awards and Grants

2024–2026 **PI, LIGEA I–III, IS CRA Class-C, CINECA Leonardo.**  
Competitive computing-time allocations:  $3 \times 100,000$  core-hours.

2026 **Danish Data Science Academy Visit Grant (1.7k EUR)**, Niels Bohr Institute (Center of Gravity).

2024 **Tacchini Prize (2k EUR) – best PhD thesis (national)**, Società Astronomica Italiana.

2022 **Marco Polo mobility grant (3.5k EUR)**, University of Bologna.

2021 **Best MSc thesis prize**, Department of Physics and Astronomy “Augusto Righi”, University of Bologna.

2021 **Best poster prize**, ISAPP Summer School on Gravitational Waves.

## Research

### Collaborations and leadership

2022–present Euclid Consortium, SWG Gravitational Waves – **co-lead, WP1 Dark sirens**

2025–present Wide-field Spectroscopic Telescope – **co-lead, WP2.2.6 Cosmology: additional probes**

2025–present LIGO-Virgo-KAGRA Collaboration, Cosmology working group

2022–present Einstein Telescope Collaboration, OSB Div. 2 Cosmology

### Visits

06/2026 **Cosmology with strong gravitational wave lensing**, Center of Gravity, Niels Bohr Institute, København (DK), with Jose Maria EZQUIAGA; funded by a DDSA Visit Grant.

04–06/2022 **Statistical methods for cosmology with gravitational waves and galaxy catalogs**, Département de Physique Théorique, Université de Genève (CH), with Michele MAGGIORE & Michele MANCARELLA.

## Scientific service

- 2023–present **Referee** for Astronomy & Astrophysics (A&A), Astrophysical Journal (ApJ), Monthly Notices of the Royal Astronomical Society (MNRAS), Journal of Cosmology and Astroparticle Physics (JCAP), and Physical Review D (APS)
- 2024–present Local organizer, GW-BO Meetings, [gw-bo.github.io](https://gw-bo.github.io)
- 2025 Local Organization Committee – XV Einstein Telescope Symposium Bologna, [link](#)
- 2024 Local Organization Committee – Galaxies & AGN with the First Euclid data and beyond, [link](#)

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## Publications ORCID SCIX

Refereed: **15** (2845 citations), of which **9** with <10 authors (520 citations). h-index: **8** (SciX, August 28, 2026).

### First-author publications

- 2026 **Nicola Borghi**, Michele Moresco, Matteo Tagliazucchi, and Giulia Cuomo. Echoes from the Dark: Galaxy Catalog Incompleteness in Standard Siren Cosmology. *A&A*, volume 706, page A199, 2026. DOI: 10.1051/0004-6361/202557354.
- 2024 **Nicola Borghi**, Michele Mancarella, Michele Moresco, Matteo Tagliazucchi, Francesco Iacovelli, Andrea Cimatti, and Michele Maggiore. Cosmology and Astrophysics with Standard Sirens and Galaxy Catalogs in View of Future Gravitational Wave Observations. *ApJ*, volume 964, page 191, 2024. DOI: 10.3847/1538-4357/ad20eb.
- 2022 **Nicola Borghi**, Michele Moresco, Andrea Cimatti, Alexandre Huchet, Salvatore Quai, and Lucia Pozzetti. Toward a Better Understanding of Cosmic Chronometers: Stellar Population Properties of Passive Galaxies at Intermediate Redshift. *ApJ*, volume 927, page 164, 2022. DOI: 10.3847/1538-4357/ac3240.
- 2022 **Nicola Borghi**, Michele Moresco, and Andrea Cimatti. Toward a Better Understanding of Cosmic Chronometers: A New Measurement of  $H(z)$  at  $z \sim 0.7$ . *ApJ Letters*, volume 928, page L4, 2022. DOI: 10.3847/2041-8213/ac3fb2.

### Publications with fewer than 10 authors

- 2026 Matteo Tagliazucchi, Michele Moresco, **Nicola Borghi**, and Chiara Ciapetti. Mind the peak: Improving cosmological constraints from GWTC-4.0 spectral sirens using semiparametric mass models. *A&A*, volume 709, page A197, 2026. DOI: 10.1051/0004-6361/202558756.
- 2026 Matteo Tagliazucchi et al. (incl. **Nicola Borghi**). Pushing spectral siren cosmology into the third-generation era: A blinded mock data challenge. *A&A*, volume 711, page A120, 2026. DOI: 10.1051/0004-6361/202659517.
- 2025 Matteo Tagliazucchi, Michele Moresco, **Nicola Borghi**, and Manfred Fiebig. Accelerating the standard siren method: Improved constraints on modified gravitational-wave propagation with future data. *A&A*, volume 702, page A244, 2025. DOI: 10.1051/0004-6361/202554827.
- 2023 Elena Tomasetti, Michele Moresco, **Nicola Borghi**, Kang Jiao, Andrea Cimatti, Lucia Pozzetti, Adam C. Carnall, Ross J. McLure, and L. Pentericci. A new measurement of the expansion history of the Universe at  $z = 1.26$  with cosmic chronometers in VANDELS. *A&A*. EDP Sciences, 2023. DOI: 10.1051/0004-6361/202346992.
- 2023 Kang Jiao, **Nicola Borghi**, Michele Moresco, and Tong-Jie Zhang. New Observational  $H(z)$  Data from Full-spectrum Fitting of Cosmic Chronometers in the LEGA-C Survey. *ApJS*, volume 265, page 48, 2023. DOI: 10.3847/1538-4365/acbc77.

### Other selected publications

- 2026 LIGO-Virgo-KAGRA Collaboration. Abac et al. (incl. **Nicola Borghi**). GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation. 5 2026.
- 2026 ET Collaboration. Abac et al. (incl. **Nicola Borghi**). The Science of the Einstein Telescope. *JCAP*, volume 03, page 081, 2026. DOI: 10.1088/1475-7516/2026/03/081.

- 2025 **Nicola Borghi**, Michele Moresco, and Richard I. et al. Anderson. Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys [white paper for the ESO Expanding Horizons call]. 12 2025.
- 2025 CosmoVerse Network. Eleonora Di Valentino et al. (incl. **Nicola Borghi**). The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. *Phys. Dark Univ.*, volume 49, page 101965, 2025. DOI: 10.1016/j.dark.2025.101965.
- 2022 Elcio Abdalla et al. (incl. **Nicola Borghi**). Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies. *JHEAp*, volume 34, pages 49–211, 2022. DOI: 10.1016/j.jheap.2022.04.002.

## Talks

### Invited

- 02/2026 Overview of ET Cosmology Science, ET science workshop for early career researchers - Rome (Italy)
- 12/2025 IASF Seminar, Istituto di Astrofisica Spaziale e Fisica cosmica - Milano (Italy)
- 12/2025 DIFI Seminar, Dipartimento di Fisica, Università di Genova (Italy)
- 11/2025 CPT Seminar, Centre de Physique Théorique - Aix-Marseille University (France)
- 11/2025 OPINAS Seminar, Max Planck Institute for Extraterrestrial Physics - München (Germany)
- 3/2024 SISSA Astrophysics Colloquium - Trieste (Italy)

### Contributed

- 7/2026 EAS2026 - Future of European Astronomy - Lausanne
- 7/2026 EAS2026 - WST in the 2040s landscape: the power of synergies - Lausanne
- 5/2026 Euclid Consortium Meeting 2026 - Barcelona (remote)
- 9/2025 Shedding light on dark sirens Workshop - Leiden
- 3/2025 Wide-field Spectroscopic Telescope - Surveying the Universe in the 2040's and beyond - Napoli
- 9/2024 110° Congresso della Società Italiana di Fisica - Bologna
- 6/2024 LXV Congresso della Società Astronomica Italiana - Napoli
- 5/2023 XIII Einstein Telescope Symposium - Cagliari
- 9/2022 International conference PUMA22 - Sestri Levante
- 7/2022 EAS2022 - Galaxies as cosmological tracers - Valencia
- 7/2022 EAS2022 - ESO@60: A stairway to the Universe - Valencia
- 4/2022 Hypatia Colloquium 2022: Early Career Astronomer series at ESO

## Supervision and Teaching

### Supervision of Visiting Students.

- 2026 L. Ceyte (from Université Paris-Saclay), *Cosmic chronometers with James Webb data*

### Co-supervision of Master's Theses, University of Bologna.

- 2026 C. Ciapetti, *Standard siren cosmology with weakly parametric BBH mass functions*
- 2025 M. Mori, *Trade-off analysis for future LVK dark sirens*
- 2025 G. Cuomo, *Astrophysical and cosmological constraints with LVK/GWTC-3 data*
- 2025 M. Friebig, *Forecasting constraints on modified gravity with upcoming LVK observing runs*
- 2024 N. Passaleva, *Enhancing the potential of gravitational waves as standard sirens: a statistical analysis*
- 2022 E. Tomasetti, *Vincoli sulla storia di espansione dell'Universo tramite cronometri cosmici nella survey VANDELS*
- 2022 M. Schulz, *Gravitational Waves as Dark Sirens: an Astrophysical and Cosmological Analysis*

2025–2026 **Teaching assistant, Elements of Informatics (INF/01), Astronomy, University of Bologna.**

2024–2025 **Teaching assistant, Elements of Informatics (INF/01), Astronomy, University of Bologna.**

- 2022–2023 **Teaching assistant, Elements of Informatics (INF/01)**, *Astronomy*, University of Bologna.
- 2020–2021 **Teaching assistant, Astrophysics Laboratory (FIS/05, optical/near-IR module)**, *Astrophysics and Cosmology*, University of Bologna.

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## Software

- Main developer:
- **CHIMERA** ([github.com/CosmoStatGW/CHIMERA/](https://github.com/CosmoStatGW/CHIMERA/)): Python code to perform gravitational wave cosmology with standard sirens and galaxy catalogs based on the Hierarchical Bayesian formalism.
  - **PyLick** ([pylick.readthedocs.io](https://pylick.readthedocs.io)): Python tool to measure spectral absorption features on galaxy spectra.
  - **cornetto** ([cornetto.readthedocs.io](https://cornetto.readthedocs.io)): Python tool to quickly generate smooth corner plots for MCMC chains.

Languages: Python (*advanced*), Julia, FORTRAN, C++, R

Other: High-performance and cloud computing, large language models, Linux/Unix, L<sup>A</sup>T<sub>E</sub>X, HTML5 & CSS, graphic design, microcontroller and single-board-computer programming (Arduino, Raspberry Pi).

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## Outreach

### Public engagement

- 2016–present **Public lectures and stargazing nights**, *Gruppo Astrofili Persicetani & Museo del Cielo e della Terra*, San Giovanni in Persiceto (BO) and surrounding areas, (50+ events, topics: [nicoborghi.github.io/#outreach](https://nicoborghi.github.io/#outreach)).
- 2017–2023 **Laboratory activities for high schools**, *Museo del Cielo e della Terra - FisicLab*, Agen.Ter., San Giovanni in Persiceto (BO).

### Press, media, and books

- 2024 **Giornale di Astronomia**, *Onde gravitazionali e cronometri cosmici per ricostruire la storia di espansione dell'Universo*, Article, 2024, vol. 50, n. 4, p. 9.
- 2022 **Media INAF**, *Cronometri cosmici per la costante di Hubble*, [Article](#), 08/04/2022.

### Books

- 2023 Nicola Borghi, Marco Cacciari, Thomas Mazzi, Romano Serra, Sandro Zannarini, *Meteoriti Storiche, un metodo per indagare il passato: Il caso Renazzo*, In riva edizioni, Bologna, 978-88-9364-439-8, [link](#).

### University's "Third-Mission"

- 2021–2026 **Fisica Experience museum**, *Scientific advisor for the cosmology section (with N. Semprini)*, [fisicaexperience.it](https://fisicaexperience.it).
- 2019–2026 **European Researchers' Night**, *Le Meraviglie del Tempo e dello Spazio (with R. Serra)*.
- 2021–2026 **Officina Laboratorio**, *Earth's motions and the concept of time*, italian PCTO program (secondary-school).
- 2021–2022 **Podcast**, *Dottorato et al.*, role: post-production, [listen on Spotify](#).

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## Other Activities

- 2024–present **Elite National Commissaire**, *Italian Cycling Federation*, (national since 2022, regional since 2017).
- 2016–present **Volunteer amateur astronomer**, *Gruppo Astrofili Persicetani*, [www.gapers.it](https://www.gapers.it).

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## Languages

- ENGLISH: Fluent (C1)
- FRENCH: Basic user
- EMILIANO (ISO 639-3: EGL): Native speaker
- ITALIANO: Native speaker